

1. You flip 11 fair coins and your friend flips 10 fair coins, find the probability that you get more heads than your friend.

Solution: Intuitively, for the first 10 times A has the same expected number of heads as B. So if the 11th flips of A results in H, then they get more heads than B, and this happens with 50% probability.

More formally, define X_A as the number of H for player A and X_B the number of H for player B. Define a variable Y_B as the number of tails obtained by B. The probability we want to compute can be expressed as

$$\Pr(X_A > X_B) = \Pr(X_A > Y_B) = \Pr(X_A > 10 - X_B) = \Pr(X_A + X_B > 10).$$

But $X_A + X_B$ is the number of H in 21 tosses (10 for B, 11 for A) of a fair coin, and this expectation is 10.5. By symmetry, the probability that this total exceeds 10 is 0.5. But you can also use the binomial distribution

`1-pbinom(10,21,0.5)`

which gives again 0.5.

2. A satellite system consists of 10 components and functions if at least 9 of the 10 components function. Assume each of the components independently functions with probability 0.9. What is the probability that the satellite system will function? (Round to 3 decimal places.)

Solution: The probability that the satellite system will function is

$$\Pr(X \geq 9) = \binom{10}{9} 0.9^9 0.1^1 + \binom{10}{10} 0.9^{10} 0.1^0 = 0.7360989$$

`dbinom(9,10,0.9) + dbinom(10,10,0.9)`

3. Suppose that $P(A)$ and $P(B)$. What does $P(A|B)$ equal if A and B are mutually exclusive?

Solution: We have that, if A and B are mutually exclusive, then $P(A \cap B) = 0$. Then

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = 0.$$

4. An urn contains two type A coins and one type B coin. When a type A coin is flipped, it comes up heads with probability $1/4$, whereas when a type B coin is flipped, it comes up heads with probability $3/4$. One coin is randomly chosen from the urn and flipped. Given that the flip landed on heads, what is the probability that it was a type A coin?

Solution:

$$\begin{aligned} P(A|H) &= \frac{P(H|A)P(A)}{P(H|A)P(A) + P(H|B)P(B)} \\ &= \frac{\frac{1}{4} \cdot \frac{2}{3}}{\frac{1}{4} \cdot \frac{2}{3} + \frac{3}{4} \cdot \frac{1}{3}} = \frac{\frac{2}{12}}{\frac{2+3}{12}} = \frac{2}{5}. \end{aligned}$$

5. From a group of 6 men and 8 women, a committee consisting of 3 men and 3 women is to be formed. How many different committees are possible if 1 man and 1 woman refuse to serve together?

Solution: In the pair that does not want to work together, call the man M and the woman W. We need to find the number of ways in which the committee need to be formed if M and W refuse to serve together.

If M is selected and W is not, we need 2 more men from the remaining 5 and since W is not selected we will have 7 women from which to select 3, i.e.

$$\frac{7}{3} \times \frac{5}{2} = 350.$$

If W is selected and M is not, we need 2 more women from the remaining 7 and since M is not selected we will have 5 men from which to select 3, i.e.

$$\frac{7}{2} \times \frac{5}{3} = 210.$$

If neither W nor M are selected, we need 3 women from the remaining 7 and 3 men from the remaining 5, i.e.

$$\frac{7}{3} \times \frac{5}{3} = 350.$$

We need to sum the three numbers together: $350 + 210 + 350 = 910$.

6. The discrete random variable X takes values 1, 2, and 4 with probabilities

$$P(X = 1) = 0.5$$

$$P(X = 2) = 0.3$$

$$P(X = 4) = 0.2.$$

What is $E(X)$, the expected value of X ?

Solution:

$$1 \cdot 0.5 + 2 \cdot 0.3 + 4 \cdot 0.2 = 1.9.$$

7. The discrete random variable X takes values 0.5, 1, and 2, and has cumulative distribution function $F_X(x) = \frac{x^2}{4}$. What is $P(X = 1)$?

Solution:

$$F(0.5) = \frac{\left(\frac{1}{2}\right)^2}{4} = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$$

$$F(1) = \frac{1^2}{4} = \frac{1}{4} P(X = 1) = F(1) - F(0.5) = \frac{1}{4} - \frac{1}{16} = \frac{3}{16}.$$

8. An urn contains 5 red and 5 blue marbles. Two marbles are withdrawn randomly (without replacement). If they are the same colour, then you win \$1.10; if they are different colours, then you win -\$1.00 (that is, you lose \$1.00). Calculate the expected value of the amount you win.

Solution: Let X denote a random variable defined as the number of red marbles chosen. Then

$$P(X = 0) = \frac{5}{10} \cdot \frac{4}{9} = \frac{2}{9}$$

$$P(X = 1) = \Pr(\text{red, blue}) + \Pr(\text{blue, red}) = 2 \cdot \frac{5}{10} \cdot \frac{5}{9} = \frac{5}{9}$$

$$P(X = 2) = \frac{5}{10} \cdot \frac{4}{9} = \frac{2}{9}$$

Then

$$E(X) = 1.10 \cdot \frac{4}{9} - 1.00 \frac{5}{9} = -\frac{6}{9} = -0.6666667.$$

9. A poker hand consists of 5 cards (in a deck of 52 cards). If the cards have distinct consecutive values and are not all of the same suit, we say that the hand is a straight. For instance, a hand consisting of the five of spades, six of spades, seven of spades, eight of spades, and nine of hearts is a straight. What is the probability that one is dealt a straight? (Use three decimal places).

Solutions:

There are $\binom{52}{5}$ possible outcomes (a total of 52 cards, and we pick 5).

There are 10 ways to choose any card of a suit, so that we can get 5 consecutive cards, depending on this chosen card, because a ace can be above the king or below the 2, but not both in the same hand: A2345 is a straight, 10JQKA is a straight, but KA234 is not a straight.

There are $4^5 - 4$ ways of choosing a poker hand, for a particular set of consecutive cards because there are 4 possible suits (4^5), but we remove the case where all cards are from the same suit.

$$10 \cdot \frac{4^5 - 4}{\binom{52}{5}} = 0.003924647$$